MATHEMATICS

LOGARITHM

THEORY SHEET

ENGLISH

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1. EXPONENTIAL FORM

$N = a^x$

N = number, a = base, x = power/index/exponent

$N > 0,\ a > 0,\ a \neq 1,\ x \in R$

Q.1 If $2^x = 8$ then find x.

Ans 3

Sol $2^x = 2^3 \Rightarrow x = 3$

Q.2 If $16^x = 32$ then find x.

Ans Sol $2^{4x} = 2^5 \Rightarrow x = 5/4$

Properties

1. $a^x \cdot a^y = a^{x+y}$ 2. $\frac{a^x}{a^y} = a^{x-y}$

3. $\left(a^x\right)^y = a^{xy}$ 4. $\frac{1}{a^x} = a^{-x}$

5. $\sqrt[n]{a}$ = n^{th} root of $a = (a)^{1/n}$, $n \in N, n \geq 2$

For eg $a^{\frac{1}{5}} = \sqrt[5]{a}$; $a^{\frac{4}{3}} \neq \sqrt[3]{\sqrt[4]{a}}$

2.1 LOGARITHMIC FORM

$N = a^x \Leftrightarrow \log_a N = x$ ($\log_a N$ read as log N to the base a)

$N > 0,\ a > 0,\ a \neq 1$

Q.1 $f(x) = \log_{(x-2)}(5-x)$ Find values of x for which f(x) is defined?

Ans $x \in (2, 5) - \{3\}$

Sol $5 - x > 0$, $x - 2 > 0$, $x - 2 \neq 1$

$x < 5$ $x > 2$ $x \neq 3$

$x \in (2, 5) - \{3\}$

Q.2 $\log_2(8) = x$, Find x

Ans 3

Sol $8 = 2^x \Rightarrow x = 3$

Q.3 $\log_{32}16 = x$, Find x

Ans 4/5

Sol $16 = 32^x$

$2^4 = 2^{5x} \Rightarrow x = 4/5$

Important points

1. $\log_N N = 1$
2. $\log_N\left(\frac{1}{N}\right) = -1$
3. $\log_a 1 = 0$
4. $\log_a(a^p) = p$
5. $a^{\log_a N} = N$ This is known as Fundamental Log Identity

Q.4 $2^x = 3$ Find x

Ans $x = \log_2 3$

Sol $2^x = 2^{\log_2 3}$

$x = \log_2 3$

2.2 PROPERTIES OF LOGARITHM

1. $\log_a m - \log_a n = \log_a(m/n)$
2. $\log_a m + \log_a n = \log_a(mn)$
3. $\log_a(m)^p = p\log_a m$
4. $\log_{a^p} m = \frac{1}{p}\log_a m$

Ex.1 $\log_2(x-1) = 3$, Find x

Sol $(x-1) = 2^3 = 8 \Rightarrow \quad x = 9$

Ex.2 $\log_{10}a + \log_{10}b - \log_{10}c = 2$, find $\frac{ab}{c}$.

Sol. $\log_{10}\left(\frac{ab}{c}\right) = 2 \Rightarrow \quad \frac{ab}{c} = 10^2 = 100$

Ex.3 $\log_{10}a - \log_{100}b = 3$, find $\frac{a}{\sqrt{b}}$

Sol. $\log_{10} a - \frac{1}{2}\log_{10} b = 3 \Rightarrow \quad \log_{10} a - \log_{10}\sqrt{b} = 3 \Rightarrow \quad \frac{a}{\sqrt{b}} = 10^3 = 1000$

2.3 BASE CHANGING THEOREM

1. $\dfrac{\log_c a}{\log_c b} = \log_b a$ 2. $\log_c a = (\log_b a)(\log_c b)$

Ex.1 $\dfrac{\log_2 25}{\log_2 5} = \log_5 25 = 2$

Ex.2 $\log_3 7 = \dfrac{\log_{10} 7}{\log_{10} 3}$

Ex.3 $(\log_2 3)(\log_3 4)(\log_4 5) = \log_2 5$

Put $c = a$ in relation (1)

3. $\dfrac{1}{\log_a b} = \log_b a$ 4. $a^{\log_c b} = b^{\log_c a}$

Ex.4 $49^{\log_7 5} = 5^{\log_7 49} = 5^2 = 25$

Note: $a^{k\log_c b} = b^{k\log_c a}$

Ex.5 $25^{2\log_5 3} = 3^{2\log_5 25} = 3^{2(2)} = 3^4 = 81$

Proof of relation (4)

$\log_c b = (\log_a b)(\log_c a)$

LHS $= a^{\log_c b} = a^{(\log_a b)(\log_c a)} = \left(a^{\log_a b}\right)^{\log_c a}$ $\left(\because a^{xy} = \left(a^x\right)^y\right)$

$= (b)^{\log_c a} = \text{RHS}$

Note:

1. If number and base of a log are on same side of unity then value of that log will be positive.
2. If number and base of a log are on different sides of unity then value of that log will be negative.

Example: $\log_7 4 = +ve$ $\log_{2/3} 5 = -ve$

$\log_{2/3} 5/7 = +ve$

3.1 ANTILOGARITHM

$N = a^x \Leftrightarrow \log_a(N) = x \Leftrightarrow \text{antilog}_a(x) = N$

For Eg. $\text{antilog}_4(3) = 4^3 = 64$

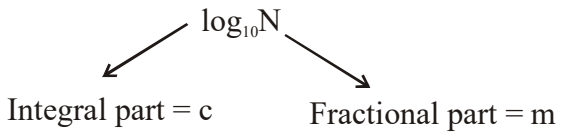
3.2 INTEGRAL PART AND FRACTIONAL PART

Number	Fractional part	Integral part
7.9	0.9	7
2.3	0.3	2
5	0	5
–2.6	0.4	–3
–5.9	0.1	–6
π	$\pi-3$	3

3.3 CHARACTERISTIC AND MANTISSA

Given a number N, Logarithms can be expressed as $\log_{10}N$ =Integer + fraction (+ve)

Integral Part = Characteristics and fractional part = Mantissa



Ex.1 $10^{3.4}$ $\log_{10}(10^{3.4}) = 3.4$ So $c = 3$ $m = 0.4$

Ex.2 2^{100}

$\log_{10}2^{100} = 100\log_{10}2$

$= 100(0.301) = 30.10$

$c = 30,$ $m = 0.10$

$\log_{10}2 = 0.3010$
$\log_{10}3 = 0.4771$
$\log_{10}7 = 0.8451$

Que. Find Characeteristic and Mantissa with respect to base 10

1. 6^{100}

Ans $c = 77, m = 0.81$

Sol $\log_{10}(6^{100})$

$= 100[\log_{10}2+\log_{10}3] = 77.81 \Rightarrow c = 77, m = 0.81$

2. 5^{50}

Ans c = 34, m = 0.95

Sol $\log_{10}5^{50} = 50\log_{10}5$

$= 50(0.6990) = 34.95 \Rightarrow c = 34, m = 0.95$

3. $3^{20}.2^{10}$

Ans c = 12, m = 0.5520

Sol $\log_{10}3^{20}.2^{10} = 20\log_{10}3 + 10\log_{10}2$

$= 12.5520 \Rightarrow c = 12, m = 0.5520$

4. 5^{-100}

c = –70, m = 0.10

Sol $\log_{10}(5^{-100}) = -100\log_{10}5$

$= -100(0.6990) = -69.90 \Rightarrow c = -70, m = 0.10$

5. $(2.5)^{-100}$

Ans c = –40 and m = 0.20

Sol $\log_{10}(2.5)^{-100} = -100[\log_{10}5 - \log_{10}2]$

$= -100(0.3980) = -39.80$

So, c = –40 and m = 0.20

6. 2^{-50}

Ans c = –16, m = 0.95

Sol $\log_{10}2^{-50} = -50\log_{10}2$

$= -50(0.3010) = -15.05 \Rightarrow c = -16, m = 0.95$

3.4 NUMBER OF DIGITS AND NUMBER OF ZEROES

Scientific notation

A positive number can be expressed as $p \times 10^n$, where $n \in I$, $p \in [1, 10)$

$2361 = 2.361 \times 10^3$; $542 = 5.42 \times 10^2$; $0.00034 = 3.4 \times 10^{-4}$; $0.00004 = 4 \times 10^{-5}$

No. of digits = Characteristic + 1

Q.1 Find number of digits in 2^{100}

Ans c = 30, D = 30+1 = 31

Q.2 Find number of digits in 6^{100}

Ans c = 77; N.D. = 78

Q.3 Find number of digits in 5^{50}

Ans $c = 34$; N.D. $= 35$

No. of zeroes $= |\text{Characteristic} + 1|$

Q.4 Find number of zeroes after decimal in 5^{-100}

Ans $c = -70 \Rightarrow$ N.Z. $= 69$

Q.5 Find number of zeroes after decimal $(2.5)^{-100}$

Ans $c = -40 \Rightarrow$ N.Z. $= 39$

Q.6 Find number of zeroes after decimal 2^{-50}

Ans $c = -16 \Rightarrow$ N.Z. $= 15$

4.1 LOGARITHMIC EQUATIONS TYPE - 1

$\log_a m = p \Rightarrow m = a^p$

$\log_a m = \log_a N \Rightarrow m = n$

Q.1 $\log_2(x+1) + \log_2(x-1) = 3$

Ans 3

Sol $x^2 - 1 = 8$

$x = \pm 3 \Rightarrow x = -3$ (Rejected)

so, $x = 3$

Q.2 $\log_5(1+\log_4 x) = 1$

Ans 256

Sol $1 + \log_4 x = 5 \Rightarrow \log_4 x = 4 \Rightarrow x = 4^4$

Q.3 $\log_4(2\log_3(1 + \log_2(1 + 3\log_3 x))) = \frac{1}{2}$

Ans 3

Sol $2\log_3(1 + \log_2(1+3\log_3 x)) = 2$

$1 + \log_2(1 + 3\log_3 x) = 3$

$\log_2(1 + 3\log_3 x) = 2$

$3\log_3 x = 3$

$x = 3$

Q.4 $1 - \log_{10}5 = \frac{1}{3}(\log_{10}\frac{1}{2} + \log_{10}x + \frac{1}{3}\log_{10}5)$

Ans $\frac{16}{5^{\frac{1}{3}}}$

Sol $3 - \log_{10}5^3 = \log\frac{x}{2} + \frac{1}{3}\log 5 = \log_{10}10^3 - \log_{10}5^3$

$\frac{10^3}{5^3.5^{\frac{1}{3}}} = \frac{x}{2}$ $\quad x = \frac{16}{5^{\frac{1}{3}}}$

Q.5 $9^{\log_3(1-2x)} = 5x^2 - 5$

Ans $-2 - \sqrt{10}$

Sol $3^{\log_3(1-2x)^2} = 5x^2 - 5$

$1 + 4x^2 - 4x = 5x^2 - 5$

$x^2 + 4x - 6 = 0$

$x = \frac{-4 \pm \sqrt{40}}{2} = -2 + \sqrt{10}, -2 - \sqrt{10}$

since $1 - 2x > 0$

$x < \frac{1}{2}$

$\therefore x = -2 - \sqrt{10}$

4.2 LOGARITHMIC EQUATIONS TYPE - 2

Based on Substitution

Q.1 $(\log_3 x)^2 - 3\log_3 x + 2 = 0$

Ans 3,9

Sol $\log_3 x = t$

$t^2 - 3t + 2 = 0$

$t = 1, 2$

$\log_3 x = 1, 2$

$x = 3, 9$

Q.2 $2^{2x} + 2^x - 2 = 0$

Ans 0

Sol $2^x = t$

$t^2 + t - 2 = 0$

$(t + 2)(t - 1) = 0$

$t = 1, -2$

$2^x = 1 \Rightarrow x = 0$

Q.3 $2^{1+x} + 2^{2-x} = 6$

Ans 0,1

Sol $2.2^x + \frac{4}{2^x} = 6$

$2t + \frac{4}{t} = 6$

$t^2 - 3t + 2 = 0$

$t = 1, 2 \Rightarrow x = 0, 1$

Q.4 $\log_2(9 - 2^x) = 3 - x$

Ans 0,3

Sol. $9 - 2^x = \frac{2^3}{2^x}$

$2^x = t$

$9t - t^2 = 8$

$t^2 - 9t + 8 = 0$

$(t - 1)(t - 8) = 0$

$t = 1, 8 \Rightarrow x = 0, 3$

Q.5 $(\log_3 x)^2 + \log_3 9x = 8$

Ans $9, \frac{1}{27}$

Sol $(\log_3 x)^2 + \log_3 9 + \log_3 x = 8$

$\log_3 x = t$

$t^2 + t - 6 = 0$

$(t + 3)(t - 2) = 0$

$t = 2, -3$

$\log_3 x = 2$ $\log_3 x = -3$

$x = 9$ $x = \frac{1}{27}$

Q.6 $(\log_2 4x)^2 + \log_2 2x = 5$

Ans $1, \frac{1}{32}$

Sol $(2 + \log_2 x)^2 + 1 + \log_2 x = 5$

$4 + (\log_2 x)^2 + 4\log_2 x + 1 + \log_2 x = 5$

$(\log_2 x)^2 + 5\log_2 x = 0$

$t^2 + 5t = 0$ $\log_2 x = t$

$t = 0, -5$

$x = 1, \frac{1}{32}$

4.3 LOGARITHMIC EQUATIONS TYPE - 3

Equations of the form $(Variable)^{variable}$

Q.1 $x^{\log_{10} x} = \frac{100}{x}$

Ans $x = 10, \frac{1}{100}$

Sol $\log_{10}\left(x^{\log_{10}x}\right) = \log_{10}\left(\frac{100}{x}\right)$

$(\log_{10}x)(\log_{10}x) = \log_{10}100 - \log_{10}x$

Put $\log_{10}x = t$

$t^2 = 2 - t$

$t^2 + t - 2 = 0$

$(t+2)(t-1) = 0 \Rightarrow t = 1, -2 \Rightarrow x = 10, \frac{1}{100}$

Note:

1. Special case of (Variable)

$a^b = a^c$

$a = 1$ or $b = c$

Proof:

$a^b = a^c$

$\log(a^b) = \log(a^c)$

$b \log a = c \log a$

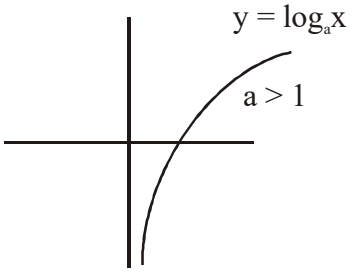
$(b - c) \log a = 0$

$b - c = 0$ or $\log a = 0$

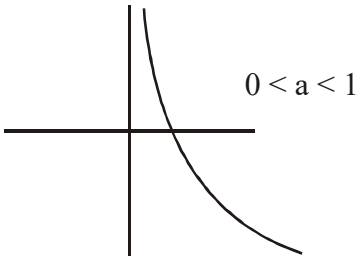
$b = c$ and $a = 1$

2. $\log_e x = \ln(x) = \log x$

5. LOGARITHMIC INEQUALITIES



f is increasing



f is decreasing

$f(x_1) > f(x_2)$ $\qquad$ $f(x_1) > f(x_2)$

$x_1 > x_2$ $\qquad$ $x_1 < x_2$

$\log_a M > \log_a N$

$a > 1$ $\qquad$ $0 < a < 1$

$M > N > 0$ $\qquad$ $0 < M < N$

$\log_a M > P$

$a > 1$ $\qquad$ $0 < a < 1$

$M > a^P > 0$ $\qquad$ $0 < M < a^P$

Q.1 $\log_2(x-1) < 3$

Ans. $x \in (1,9)$

Sol $0 < x - 1 < 8$

$1 < x < 9$

Q.2 $\log_{\frac{1}{2}}\left(\frac{4x-1}{x+7}\right) > -1$

Ans $x \in (1/4, 15/2)$

Sol $0 < \frac{4x-1}{x+7} < \left(\frac{1}{2}\right)^{-1}$ $\qquad$ $\frac{4x-1}{x+7} > 0$

$\frac{4x-1}{x+7} < 2$ $\qquad$ $x \in (-\infty,-7) \cup \left(\frac{1}{4},\infty\right)$

$\frac{4x-1}{x+7} - 2 < 0$

$\frac{2x-15}{x+7} < 0$

$x \in \left(-7, \frac{15}{2}\right)$

Final Answer = $x \in \left(\frac{1}{4}, \frac{15}{2}\right)$

In case, base is variable

Q.3 $\log_{(x+3)}(x^2 - x) < 1$

Ans. $x \in (-3,-2) \cup (-1, 0) \cup (1,3)$

Sol. Case-I : $x + 3 > 1 \rightarrow A_1$

$A_1 = x > -2$

$0 < x^2 - x < x + 3 \rightarrow A_2$

$A_2 \Rightarrow x \in (-1,0) \cup (1,3)$

$C_1 = A_1 \cap A_2$

$C_1 \Rightarrow (-1,0) \cup (1,3)$

Case-II : $0 < x + 3 < 1 \rightarrow A_3$

$A_3 \Rightarrow x \in (-3,-2)$

$x^2 - x > x + 3 > 0 \rightarrow A_4$

$A_4 \Rightarrow x \in (-3,-1) \cup (3,\infty)$

$C_2 = A_3 \cap A_4$

$C_2 \Rightarrow x \in (-3,-2)$

Final Answer = $C_1 \cup C_2$

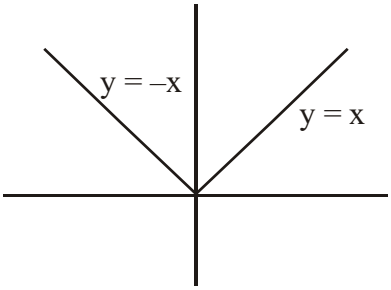
$x \in (-3,-2) \cup (-1, 0) \cup (1,3)$

6.1 MODULUS FUNCTION

$f(x) = \sqrt{x^2} = |x|$ = absolute value of x

$|3| = 3,\ |-4| = 4,,|-5| = 5,\ |\ 0\ | = 0$

$$f(x) = |x| = \begin{cases} x & if\ x \geq 0 \\ -x & if\ x < 0 \end{cases}$$



Properties of Modulus

1. $|x| \geq 0$

   Ex. $|x + 2| = -2 \Rightarrow x \in \phi$

2. $|x| = |-x|$

   Ex. $|1 - x| = |x - 1|$

3. $|x|^2 = x^2$

4. $|xy| = |x||y|$

5. $|x + y| = |x| + |y|$; if $xy \geq 0$

6. $|x - y| = |x| + |y|$; if $xy \leq 0$

6.2 MODULUS EQUATION TYPE - 1

$|x| = \text{constant} (\geq 0)$

Q.1 $|x| = 3$

Ans $x = 3$ or $x = -3$

Q.2 $\left|\frac{x+1}{x-2}\right| = 4$

Ans 3, 7/5

Sol $\frac{x+1}{x-2} = 4, \frac{x+1}{x-2} = -4$

Q.3 $|x + 5| = -1$

Ans $x \in \phi$

Q.4 $|2 + |x - 3|| = 3$

Ans 4,2

Sol $2 + |x - 3| = 3$ or $2 + |x - 3| = -3$

$|x - 3| = 1$

$x - 3 = 1$ $x - 3 = -1$

$x = 4, 2$

6.3 MODULUS EQUATION TYPE - 2

Based on $|x|^2 = x^2$

1. $x^2 - 5|x| + 4 = 0$

Ans $\pm 1, \pm 4$

Sol Put $|x| = t$

$t^2 - 5t + 4 = 0$

$t = 1,$ or $t = 4$

$|x| = 1$ $|x| = 4$

$x = \pm 1$ $x = \pm 4$

2. $|x - 3| = |2x - 1|$

Ans $x \in \left\{-2, \frac{4}{3}\right\}$

Sol on squaring,

$(x - 3)^2 = (2x - 1)^2$

$x^2 - 6x + 9 = 4x^2 - 4x + 1$

$3x^2 + 2x - 8 = 0$

$x = -2, \frac{4}{3}$

No extra solution, so $x \in \left\{-2, \frac{4}{3}\right\}$

3. $|x - 3| = 2x - 1$

Ans 4/3

Sol $(x - 3)^2 = (2x - 1)^2$

$x = -2, \frac{4}{3}$

Now, check these value

$x = -2$	$x = \frac{4}{3}$
$+ve = 2(-2)-1$	$+ve = \frac{8}{3}-1$
$(-ve)$	$(+ve)$

$$x \in \left\{\frac{4}{3}\right\}$$

4. $|x - 3| = 1 - 2x$

Ans -2

Sol $(x - 3)^2 = (1 - 2x)^2$

$x = -2, 4/3$

We will recheck now,

$x = -2$	$x = 4/3$
$+ve = 1 - 2(-2)$	$+ve = 1 - 8/3$
$(+ve)$	$(-ve)$

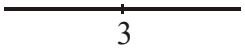
$$x \in \{-2\}$$

6.4 MODULUS EQUATION TYPE - 3

1. $|x - 3| = 2x - 1$

Ans $4/3$

Sol



Case-I

$x < 3$

$-(x - 3) = 2x - 1$

Case-II

$x \leq 3$

$x - 3 = 2x - 1$

$-x + 3 = 2x - 1$	$x = -2$
$3x = 4$	
$x = \frac{4}{3}$	for $x = -2$
for $x = 4/3$	$\lvert -2 - 3\rvert = 2\times(-2) - 1$
$\left\lvert\frac{4}{3} - 3\right\rvert = 2\times\frac{4}{3} - 1$	$\lvert -5\rvert = -5$
$\frac{5}{3} = \frac{5}{3}$	$5 \neq -5$
$x = 4/3$	$x \in \left\{\frac{4}{3}\right\}$

2. $|x - 1| + |x - 3| = 6$

Ans $\{-1, 5\}$

Sol



Case-I $x < 1$	**Case-II** $1 \le x < 3$	**Case-III** $x \ge 3$
$-(x - 1) - (x - 3) = 6$	$x - 1 - (x - 3) = 6$	$x - 1 + x - 3 = 6$
$-x + 1 - x + 3 = 6$	$x - 1 - x + 3 = 6$	$2x = 10$
$x = -1$	$2 = 6 \Rightarrow x \in \phi$	$x = 5$
	$x \in \{-1, 5\}$	

7.1 MODULUS INEQUALITIES TYPE - 1

$|x| <$ constant or $|x| >$ constant

1. $|x| < 3$

Sol $-3 < x < 3$

2. $\left\lvert\frac{x+1}{x-2}\right\rvert < 2$ $\qquad -2 < \frac{x+1}{x-2} < 2$

Sol $\frac{x+1}{x-2} > -2$ $\frac{x+1}{x-2} < 2$

$\downarrow$ $\downarrow$

A_1 A_2

$A_1 \cap A_2$

3. $|x| > 4$

Sol $x > 4$ or $x < -4$

4. $\left|\frac{x+1}{x-3}\right| > 2$

Sol $\frac{x+1}{x-3} > 2$ or $\frac{x+1}{x-3} < -2$

$\downarrow$ $\downarrow$

A_1 A_2

$A_1 \cup A_2$

5. $|x| < -3$ $x \in \phi$

6. $|x| > -3$ $x \in R$

7. $\left|\frac{x-1}{x-5}\right| > -1$

Sol $x \in R - \{5\}$

7.2 MODULUS INEQUALITIES TYPE - 2

Q.1 $|x - 3| < |2x - 1|$

Sol $(x - 3)^2 < (2x - 1)^2$

$(2x - 1)^2 - (x - 3)^2 > 0$

$(3x - 4)(x + 2) > 0$

$x \in (-\infty, -2) \cup \left(\frac{4}{3}, \infty\right)$

Q.2 $\left|\frac{x+1}{x-3}\right| > 2$

Sol $\left(\frac{x+1}{x-3}\right)^2 > 4$

$\left(\frac{x+1}{x-3}\right)^2 - 4 > 0 \qquad \Rightarrow \left(\frac{x+1}{x-3}-2\right)\left(\frac{x+1}{x-3}+2\right) > 0$

$\frac{(x-7)(3x-5)}{(x-2)(x-3)} < 0$

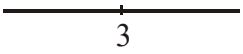
$x \in \left(\frac{5}{3}, 2\right) \cup (3, 7)$

7.3 MODULUS INEQUALITIES TYPE - 3

1. $|x-3| < 2x-1$

Ans $x \in \left(\frac{4}{3}, \infty\right)$

Sol



Case-I

$x \geq 3$

$x - 3 < 2x - 1$

$x > -2$

Case-II

$x < 3$

$-x + 3 < 2x - 1$

$3x > 4 \Rightarrow x > 4/3$

$x \in \left(\frac{4}{3}, 3\right) \cup [3, \infty)$

$x \in \left(\frac{4}{3}, \infty\right)$

2. $|x-1| + |x-3| < 6$

Ans $x \in (-1, 5)$

Sol



Case-I $x < 1$ **Case-II** $1 \leq x < 3$ **Case-III** $x \geq 3$

$-x+1-x+3<6$	$x-1-x+3<6$	$x-1+x-3<6$
$-2x<2$	$2<6$	$2x<10$
$x=-1$	$x\in[1,3)$	$x<5$
	$x\in(-1,5)$	

8. IRRATIONAL INEQUALITIES

1. $\sqrt{a}>\sqrt{b}\Rightarrow \quad a>b\geq 0$

2. $a\sqrt{b}>0\Rightarrow \quad b>0$ and $a>0$

3. $a\sqrt{b}\geq 0$

Case-I $b\geq 0$ and $a\geq 0$ Case-II $b=0$

Ex. $(x-3)\sqrt{x-1}\geq 0$

Ans. $x\in[3,\infty)\cup\{1\}$

Sol. Case-I : $x-3\geq 0$ $x-1>0$

$x\geq 3$ $x>1$

$x\in[3,\infty)$

Case-II : $x-1=0$

$x=1$

Final answer $\Rightarrow x\in[3,\infty)\cup\{1\}$

4. $a>\sqrt{b}$

Ex. $x>\sqrt{x+2}$

Ans. $x\in(2,\infty)$

Sol. $x>0$

$x^2>x+2$

$x^2-x-2>0$

$(x-2)(x+1)>0$

$x\in(-\infty,-1)\cup(2,\infty)$

Final answer = $x\in(2,\infty)$

5. $a < \sqrt{b}$

Ex. $x < \sqrt{x+2}$

Ans. $x \in [-2,2)$

Sol. Case-I : $x \le 0$

$x + 2 \ge 0$

$x \ge -2$

$x \in [-2,0]$

Case-II : $x > 0$

$x^2 < x + 2$

$x^2 - x - 2 < 0$

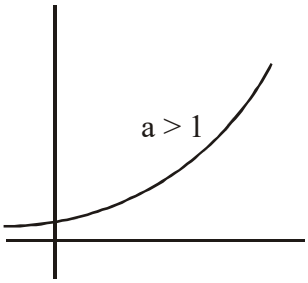
$x \in (-1,2)$

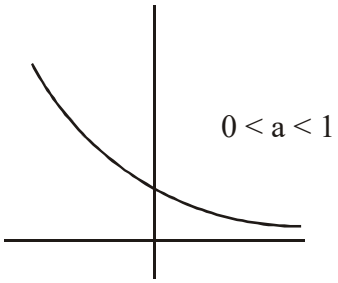
$x \in (0,2)$

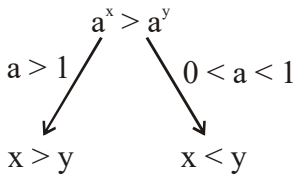
Final answer = $x \in [-2,2)$

9. EXPONENTIAL INEQUALITIES

$y = a^x$







Q.1 $2^x < 8$

Ans. $x < 3$

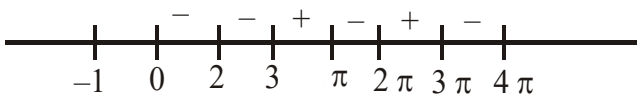
Sol $2^x < 2^3 \Rightarrow$

Q.2 $\dfrac{\left(2^x-4\right)\left(x-3\right)\left(x+1\right)}{\left(\log_2 x-1\right)\sin x} < 0$

Ans $x \in (0, 2) \cup (3, \pi) \cup (2n\pi, (2n + 1)\pi)$

Sol. $2^x - 4 = 0$, $x - 3 = 0$, $x + 1 = 0$, $\log_2 x - 1 = 0$, $\sin x = 0$

$(x = 2)$, $(x = 3)$, $(x = -1)$, $(x = 2, x > 0)$, $x = 0, \pi, 2\pi, 3\pi$.....



$\therefore x \in (0, 2) \cup (2,3) \cup (\pi, 2\pi) \cup (3\pi, 4\pi)$.....

final $x \in (0, 2) \cup (2,3) \cup (2n\pi + \pi, 2n\pi + 2\pi)$